

CHAPTER 11

Laying of Hands: Transforming a Portable Radio into a Synthesizer by Making Your Skin Part of the Circuit

You will need:

- A battery-powered radio.
- Batteries for the radio.
- Small screwdrivers, flat and/or Phillips, as required to disassemble the radio.
- Plastic electrical tape and some stranded hookup wire may be needed.
- Optional: cigar box, double-stick foam tape.

HOW TO CHOOSE A RADIO

It should be cheap enough that you won't be too angry if it never works again. The AM band is more important than the FM, but it doesn't matter if the radio picks up both. It should have *analog* tuning (i.e. a dial) rather than digital presets or scan buttons. Older radios are usually better than newer ones. Larger radios are easier to work with than tiny ones, and often produce a wider range of sounds. Boom boxes are great, and you can use the tape head for other experiments (see Chapter 9). It's better if it has a built-in speaker, not just a headphone jack, but a headphone jack *in addition to* a speaker can be useful. And most importantly: IT MUST BE BATTERY POWERED! Beware: an alarm clock radio with a built-in "backup battery" is *not* suitable, since it requires AC power to function as a radio.

LAYING OF HANDS

Install the batteries and confirm that the radio works *prior* to disassembly; if not functional, return it to the store. If it works, remove the batteries.

Remove the screws holding radio together. Put them somewhere safe (like a cup, *not* loose on top of your table), taking care to make a note of location if they are of different sizes. Some screws may be hidden beneath stickers or under the batteries. Gently

separate the halves of the radio. If plastic wedge-fasteners are used you may need to twist a thin flat screwdriver or clam shucker along the seams. Don't force it—check for hidden screws if it resists. Avoid tearing wires. Once open, make note of any wires connecting the two halves of the radio or the circuitry to the speaker, battery, antenna, etc., in case they get torn later (see the Fourth Rule of Hacking).

Locate and remove the screws holding the circuit board to the radio housing. Carefully remove the circuit board from the chassis. Sometimes adhesive may be used as well as screws. Knobs and switches may intrude into slots in the case and require bending or cutting the plastic to release the board; alternatively, you can remove the knobs (they usually attach with small screws). Circuit boards, especially cheap ones, can be very brittle, so don't bend the board!

The side of the board with most of the little bumpy colorful things (resistors, capacitors, chips, etc.) is called the "component side"; the side that consists mostly of



Figure 11.1 Laying of hands.

wiggly lines (usually silver or copper colored, sometimes under a translucent green wash) is the "solder side." Turn the board so that the solder side is accessible. If it has a telescoping antenna this may need to be disconnected in order to expose the circuit board—you probably won't need to reconnect it. Remove the volume and tuning knobs if they are large enough to cover over parts of the circuit board, leaving short nubbins by which you should still be able to adjust settings. Replace the batteries; depending on the construction of the case you may have to hold the batteries in place using plastic electrical tape, or extend the battery leads from the discarded case with extra wire.

Turn on the radio and tune it to a "dead spot" between stations or at the end of the dial. Lick your fingertips, like a safecracker in a film noir. Touch the circuit board lightly in different places with your fingers until you find a location that affects the radio's sound (see Figure 11.1). Search for touch points that cause the radio to start to whistle, squeal, or "motorboat." Tune the radio across the band, and continue to experiment with finger placement. Try several fingers at once. The moisture on your fingertips increases conductivity, and makes your touch more sensitive, but I suggest you do not lick the circuit board directly. And, observing Seventh Rule of Hacking, avoid shorting points on the circuit board with screwdriver tips, bare wire, or full immersion in drool.

Don't worry if you don't get new sounds immediately—it's a bit like trying to make your first sound on a trumpet or flute, or learning to ride a bike. Sometimes you have to work a while before you find a sweet spot, but then you'll lock in and form a tight feedback relationship with the instrument, and the sounds should pour forth (see audio track 10). It can take up to an hour to make your first squeal. If you can't get anything after an hour, try another day or another radio.

WHAT'S HAPPENING?

As Ol' Sparky has demonstrated on far too many occasions, flesh is an excellent conductor of electricity. By bridging different locations on the board with your fingers you are effectively—if haphazardly—adding free-range resistors and capacitors to the existing circuit. Your body literally becomes part of the circuit. Varying the pressure (or dampness) of your finger changes the values of these components. Depending on the location and pressure, you may end up merely re-tuning the radio, or affecting its loudness, but you may change the radio into a very different kind of circuit, like an oscillator. This happens when the output of a gain stage (such as an amplifier) flows back through your skin into an input—voilà, feedback, the musician's friend!

A radio contains most of the basic modules of a classic analog music synthesizer: oscillators, noise generators, amplifiers, filters, ring modulators—in addition to the world's largest sample library, courtesy your local radio stations. Your skin re-tunes these modules, patches them together, and adds feedback paths. Moving your fingers and changing the volume, tuning, and band selections reconfigure this synthesizer to make different sounds—a whistling oscillator, modulated white noise, a signal processor chopping fragments of radio broadcasts, etc.

You may not know exactly what you are doing (like plugging patchcords and twiddling knobs in the dark), but you should soon acquire a sense of touch: what points work best, how does pressing harder affect the sound, etc. This is a very direct,

100% 1500 PUCISA JA
TOSHIKAZU. COMO RESSON
PO 40206. 11-00 50 PER
AUDIO

THE CRACKLEBOX

Legend has it that the inspiration for the Cracklebox (*Kraakdoos*, as it is known in Dutch) springs from the out-of-body experience of an adolescent Michel Waisvisz (1949–2008) after he attempted to play his father's shortwave receiver by laying his hands on its 240 volt-powered circuit board. He recovered, the radio was nailed against the wall of their house in Delft to prevent further mishap, and Waisvisz's vision of an electronic instrument that could be played by intuitive touch was eventually safely realized in the late-1960s in collaboration with the engineer Geert Hamelberg. After building a number of different keyboard-sized instruments, in 1975 Waisvisz worked with engineers at STEIM, a music research foundation in Amsterdam, to design the very portable and affordable Cracklebox. Four thousand Crackleboxes were sold, and the original instrument was reissued a few years ago and can be bought online (www.steim.org/steim/cracklebox.php). Touch circuits had been employed in the expressive keyboard controllers of the maverick synthesizer designers Donald Buchla in 1965 and Serge Tcherepnin in the early 1970s, but the Cracklebox was the first mass-produced electronic musical instrument that incorporated the player's skin as the primary variable component in a sound-generating circuit.

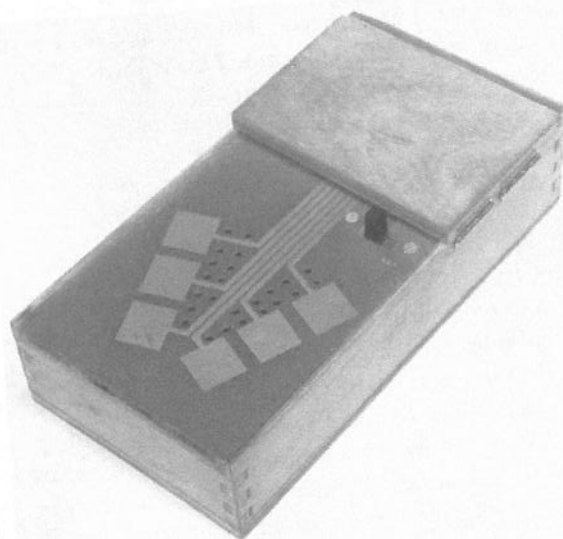


Figure 11.2
The Cracklebox.

interactive sense of control similar to that which a “real” instrumentalist, such as a violinist, uses to articulate and intonate notes. This principle of direct contact with circuitry is relatively rare among commercial electronic instruments, but has often been exploited by experimentalists (and is the soul of the infamous Cracklebox).

In the upcoming chapters we will modify circuits by replacing your flesh with specific “knowable” components—the effect may be more predictable and stable, but the sense of touch will be diminished. In the future if things start to sound too controlled, remember you can always add your body to the circuit. And if your eviscerated radio becomes too

predictable, try a friend's or open another—different radios respond differently. (Listen to the Bent Radio Orchestra in Duncan Chapman and Stewart Collinson's video on the DVD.)

Older-style radios sometimes have tuning coils whose colorful slotted tops are just asking for the twist of a screwdriver. Doing so may diminish or disable the radio's ability to pick up stations, but can add whooshy noise and rhythmic motorboating to your instrument's palette (see Michael Bullock's video on the DVD).

Generally speaking, the older the radio, the greater the range of sounds you can coax out of it. Modern radios cluster more functions onto each chip, minimizing the number of interconnections you can make with your skin. Older designs use more components on larger, finger-friendlier circuit boards; they often have bigger speakers, with more extended bass response as well. When scavenging flea markets, thrift shops and eBay, think 1960s (like the radio in the lower center of Figure 11.1), rather than twenty-first century. The additional weight and mass in your touring luggage will be more than offset by its sonic splendor.

When you are through experimenting you may want to reassemble the radio—this is the safest way to carry it around, and to insure its future functionality as a radio. But if you are so enamored of your electronic Ouija board that you cannot bear to seal it up again, welcome to the most hardware part of hacking: finding a box. Cigar boxes work great: using double-stick tape, you can stick down the circuit board (solder side up), speaker, and related parts (see Figure 11.3). Close the lid to transport, open it to play. Don't do this with metal boxes, as they may short out the circuit, but wood or plastic are fine.

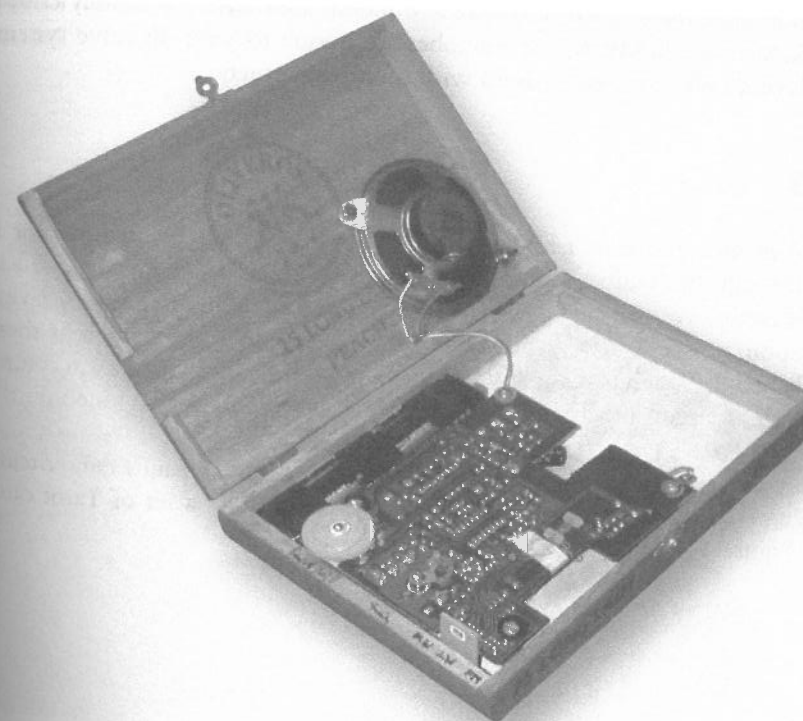


Figure 11.3 “Laying of Hands” radio mounted inside a cigar box, Nicolas Collins.